HybridPisaNegotiator: A Scale-Invariant Hybrid Bidding Agent with Bayesian Opponent Modelling and LLM-Generated Dialogue

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Abstract

We present *HybridPisaNegotiator*, an automated negotiation agent for the ANL 2026 competition built on the NegMAS framework. The agent decouples *strategy* from *language*: bidding and acceptance are governed by a classical time- and behaviour-based concession heuristic (Keskin et al., 2021), augmented by a Bayesian opponent model that estimates the partner's utility function online, while a local large language model (LLM) generates short, human-sounding messages that present the already-chosen offer. All utilities are normalised internally to $[0, 1]$, making the agent scale-invariant across structurally different domains. We describe the agent architecture and decision rules, and report an empirical evaluation across the Trade, Island, and Grocery scenarios against a range of baseline opponents.

1 Introduction

Automated bilateral negotiation under the Stacked Alternating Offers Protocol (SAOP) requires an agent to decide, at each round, whether to accept the opponent's standing offer or to propose a counter-offer. A strong agent must (i) concede at an appropriate rate given the deadline, (ii) model what the opponent values so that concessions are spent efficiently, and (iii) accept at the right moment to avoid both premature capitulation and costly time-outs. The ANL 2026 setting additionally introduces a natural-language channel: agents may exchange free-text messages alongside their offers.

Our design philosophy is *robust separation of concerns*. The strategic core (i.e. what to offer and when to accept) is fully deterministic and heuristic, so its behaviour is predictable and cannot be derailed by an unreliable language model. The LLM is confined to a purely cosmetic role which is turning an already-decided bid into a fluent message. This mirrors the architecture of strong heuristic agents while still exploiting generative AI where it is low-risk.

2 Agent Architecture

At each round the agent (1) updates its opponent model with the received offer, (2) classifies the opponent's behaviour and adapts its concession parameters, (3) computes a target utility, (4) selects the bid near that target that the opponent is most likely to accept, (5) decides whether to accept instead, and (6) attaches a natural-language message. We detail each component below.

2.1 Utility normalisation

Different domains express utilities on different scales (e.g. $[0, 1]$ for Trade but $[0, 100]$ for Island). Because every threshold in the strategy assumes a $[0, 1]$ scale, the agent normalises its utility function once at the start of the negotiation,

$$\hat{u}(\omega) = \frac{u(\omega) - u_{\min}}{u_{\max} - u_{\min}}, \tag{1}$$

and performs *all* internal reasoning with $\hat{u}$. The original utility function is left untouched for scoring. This single step makes the agent scale-invariant across domains.

2.2 Time-based concession

The time-based target utility is a quadratic Bézier curve in normalised negotiation time $t \in [0, 1]$,

$$U_{\text{time}}(t) = (1-t)^2 p_0 + 2(1-t)t\, p_1 + t^2 p_2, \tag{2}$$

with initial utility p_0, control point p_1, and floor p_2 (lower bounded by the reservation value). The control point p_1 shapes how Boulware- or Conceder-like the curve is.

2.3 Behaviour-based concession

The behaviour-based component reacts to the opponent's recent concessions. Let Δ_i be the change in the utility (to us) of consecutive opponent offers and w a recency-weighted window. With empathy parameter p_3,

$$U_{\text{beh}}(t) = u^{\text{me}}_{\text{last}} - (p_3 + p_3 t) \sum_i \Delta_i w_i. \tag{3}$$

The two components are blended so that behaviour dominates early and time pressure dominates near the deadline,

$$U_{\text{target}}(t) = (1 - t^2)\, U_{\text{beh}}(t) + t^2\, U_{\text{time}}(t). \tag{4}$$

2.4 Opponent behaviour classification and parameter adaptation

Using the variance of inter-offer similarity and utility changes, the agent labels the opponent as *competitive*, *cooperative*, or *silent*. Each label maps to target values for (p_1, p_2, p_3), towards which the live parameters are nudged each round with an exponential moving average (smoothing factor α),

$$p \leftarrow (1 - \alpha)\, p + \alpha\, p^{\text{target}}. \tag{5}$$

The EMA bounds the adaptation, so parameters converge smoothly and never drift unboundedly even in very long negotiations.

2.5 Bayesian opponent modelling

The agent feeds every received offer to a scalable Bayesian model that learns per-issue value preferences online and exposes an estimated opponent utility $\tilde{u}^{\text{opp}}(\omega) \in [0, 1]$. This estimate is used to spend concessions efficiently (Section 2.6) and also improves the competition's opponent-modelling score.

2.6 Bid selection

Bid selection is a two-stage procedure. The concession curve decides *how much* to concede (the target utility); the opponent model decides *which* bid to offer at that level. From all outcomes within a window of U_{target} the agent picks the one maximising

$$\text{score}(\omega) = 0.6\, \hat{u}(\omega) + 0.2\, \text{sim}(\omega, o_{\text{last}}) + 0.2\, \tilde{u}^{\text{opp}}(\omega), \tag{6}$$

where sim is the issue-wise similarity to the opponent's last offer. Our own utility dominates; opponent appeal breaks ties towards mutually acceptable deals.

2.7 Acceptance strategy

The agent accepts using a combination of standard acceptance criteria: *ACtarget* (the offer meets the current target), *ACnext* (the offer is at least as good as the bid we are about to send), and *ACcombi*, an end-game rule that, near the deadline, accepts any offer that is close to the best the opponent has ever made. A final deadline safeguard accepts any offer strictly above the reservation value rather than time out at zero. All criteria respect the reservation-value floor.

2.8 LLM dialogue layer

After the bid is chosen, a local LLM is asked to write one short, natural sentence presenting it, grounded in the scenario description, the issue space, the chosen offer, and the opponent's last offer. The layer is engineered so it *cannot* harm the negotiation:

- a deterministic template is always computed as a fallback;
- a one-time warm-up loads the model and tests reachability;
- a hard per-call timeout and a circuit breaker disable the LLM after repeated failures;
- outputs are sanitised to ASCII and validated before use.

If the model is slow or unavailable, the agent silently falls back to templates and negotiates exactly as before. Crucially, the LLM never influences bidding or acceptance.

3 Evaluation

4 Evaluation

We evaluate `HybridPisaNegotiator` along two axes: the quality of its negotiation outcomes and the behaviour of its dialogue layer. All experiments use the three competition scenarios, which were deliberately chosen to be structurally different: *Trade*, a small two-issue buyer/seller domain (40 outcomes) with utilities already normalized to $[0, 1]$; *Island*, a purely distributive eight-item split (256 outcomes) whose utilities are expressed on a raw 0–100 scale; and *Grocery*, the largest domain (625 outcomes).

To measure outcome quality we run a round-robin tournament in which every agent negotiates against every other agent and itself, on each scenario. The opponent pool spans a representative range of standard strategies: aggressive fixed-threshold acceptors (`SimpleNeg`), time-based conceders (`ConcederTB`, `LinearTB`, `BoulwareTB`, `AspirationNeg`), the built-in hybrid bidder (`Hybrid`), a tit-for-tat strategy (`NaiveTitForTat`), and a template-based adapter. For each ordered pairing we record the agent's own utility, normalized to $[0, 1]$ so results are comparable across the differently scaled domains, and average over both side assignments and repetitions; a failure to agree is scored at the reservation value. We then examine the resulting payoff matrices (Section 4.1) and illustrate the natural-language messages produced by the LLM layer (Section 4.2).

4.1 Tournament heatmaps

We ran a round-robin tournament in which every agent negotiates against every other agent (and itself) on each scenario. Figures 1, 2, and 3 show one heatmap per scenario. Each cell (i, j) reports the *row* agent i's own normalized utility in $[0, 1]$ when negotiating against the *column* agent j, averaged over both side assignments and over repetitions; a time-out is scored at the reservation value (0 after normalization). The matrices are therefore intentionally asymmetric: cell (i, j) is i's payoff and cell (j, i) is j's, so a single negotiation contributes to two different cells. Our agent is the `HybridPisa` row/column; reading along its row shows how much utility it secures against each opponent.

Trade (Fig. 1). On the two-issue Trade domain our agent extracts high utility from conceding opponents (e.g. 0.90 against `ConcederTB` and 0.73 against `LinearTB`) and a balanced split ($\approx$ 0.55) against the Boulware-style hold-outs (`BoulwareTB`, `TemplateAdapter`, `Aspiration`), which mirror its own slow-concession behaviour. The hardest opponents are `SimpleNeg` (0.29) and the built-in `Hybrid` (0.35): both are aggressive and accept only near-ideal offers, so agreements settle in their favour or time out. Self-play sits at a fair 0.55, as expected for two identical strategies.

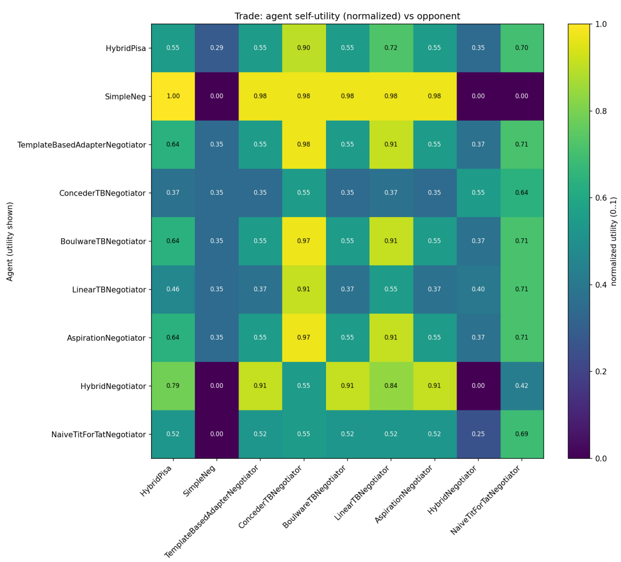

Figure 1: Trade: each agent's normalized self-utility (rows) against each opponent (columns), role- and repetition-averaged.

Island (Fig. 2). Island is a purely distributive item-split with a raw 0..100 utility scale; the internal normalization (Section 2.6) is what makes the agent behave sensibly here rather than capitulating immediately. Our agent performs strongly across the board, with 1.00 against the built-in `Hybrid`, 0.95 against `SimpleNeg`, and 0.93 against `ConcederTB` while maintaining a competitive $\approx$ 0.64 against the Boulware-class opponents. This confirms that the strategy is scale-invariant and transfers to a domain structurally different from Trade.

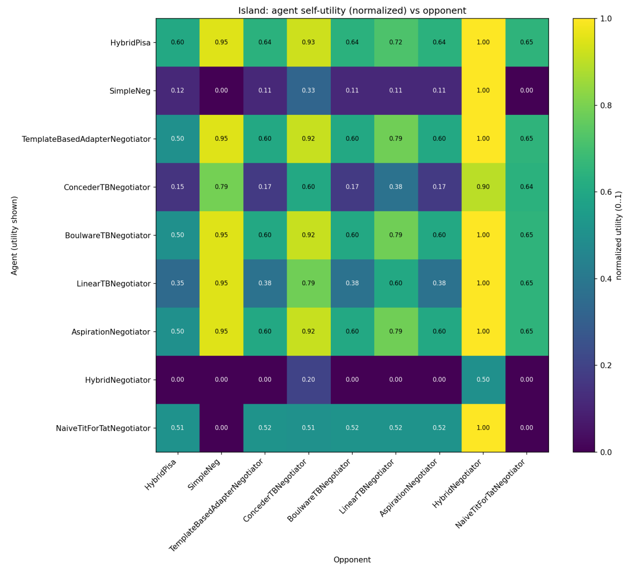

Figure 2: Island: normalized self-utility on the distributive item-split domain.

Grocery (Fig. 3). Grocery is the largest domain (625 outcomes). Here the agent is the most consistent of the three scenarios, scoring between 0.62 and 0.83 against every opponent (e.g. 0.83 vs `Hybrid`, 0.80 vs `NaiveTFT` and in self-play, 0.72 vs `SimpleNeg`). The larger outcome space gives the two-stage bid selection more room to find offers that are both high-utility for us and acceptable to the opponent, which smooths out the extremes seen on Trade.

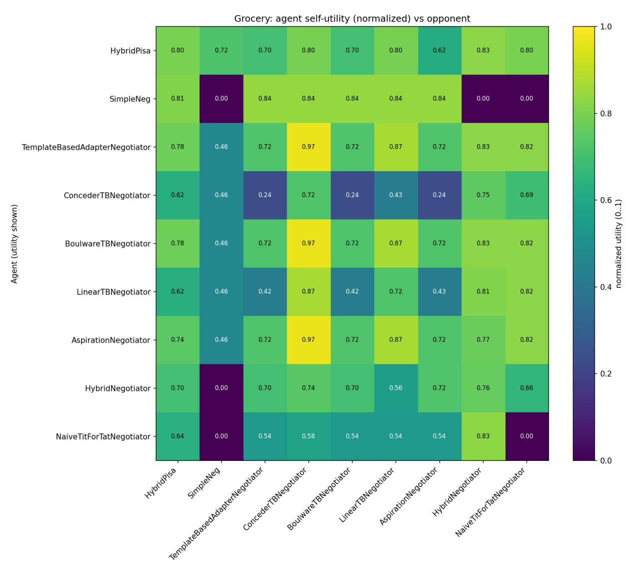

Figure 3: Grocery: normalized self-utility on the largest outcome space (625 outcomes).

Overall. Across all three domains the agent never collapses: it converts opponent concessions into high payoffs, holds a fair split against equally firm strategies, and remains robust on the un-normalized Island scale. Its weakest results are against opponents that combine aggressive bidding with a high acceptance threshold (`SimpleNeg`, built-in `Hybrid`) on the small Trade domain, where there is little room to trade concessions efficiently.

4.2 Example LLM-generated messages

To illustrate the dialogue layer, Table 1 shows sample messages generated by our agent (Qwen3-4B via Ollama) during real negotiations against a conceding opponent. The messages are grounded in the injected scenario context (scenario name, issues, and the chosen offer), so they reference the actual deal terms of each domain while the bid itself is selected by the heuristic core. When the model is slow or unavailable the agent silently falls back to fixed templates, so dialogue never affects the negotiation outcome.

Scenario	Generated message
Trade	*(opening)* "How about 5 units at 200? That feels fair — way better than 7 at 120, honestly!"
	(counter) "Hey, how about 6 units at 200? That feels fairer — more value for you, same quality, just a little more balanced."
Island	*(opening)* "Hey, how about we both go easy on it — what if Bob gets everything, and Alice just keeps the match and food? Seems fair, right?"
	(counter) "So if you're really going for that, how about I get the food and medicine, and you take the rest?"
	(accept) "Love how you've kept it all the same — perfect! I'm totally down with that, just like you."
Grocery	*(opening)* "Hey, how about we go with 4 apples, 1 banana, 4 oranges, and 4 watermelons? That feels like a great step up — just sayin'!"
	(counter) "So, how about we go with apples 4, oranges 4, watermelons 4 — just bump watermelons from 2 to 4? Seems fair, right?"

Table 1: Example messages generated by the LLM dialogue layer on each scenario. The message type (opening offer, counter-offer, acceptance) is shown in parentheses. Note how each message references the domain's actual issues: quantity and price for Trade, named items for Island, and fruit counts for Grocery, while the underlying bid is chosen by the heuristic strategy.

5 Discussion and Limitations

The agent's strengths are robustness and scale-invariance: a deterministic, well-understood strategic core that performs consistently across domains, with generative AI added only where it is safe. Limitations include: the behaviour classifier is a heuristic and can be noisy; bid selection is a linear scan, which may be costly on very large outcome spaces; and the per-issue description context for the LLM is only available when scenario metadata is co-located with the agent. Future work includes tournament-based tuning of the concession parameters and a Pareto-aware bid-selection stage.

6 Conclusion

HybridPisaNegotiator combines a classical hybrid concession strategy, a Bayesian opponent model, and a carefully sandboxed LLM dialogue layer into a single, scale-invariant negotiation agent. Its separation of strategy from language yields predictable, robust behaviour while still benefiting from natural-language interaction.

References

[1] M. O. Keskin, U. Çakan, and R. Aydoğan, "Solver Agent: Towards Emotional and Opponent-Aware Agent for Human-Robot Negotiation," in *Proc. 20th Int. Conf. on Autonomous Agents and MultiAgent Systems (AAMAS)*, 2021.

[2] Y. Mohammad et al., "NegMAS: A Platform for Automated Negotiations," *PRIMA*, 2021.